

Follow the mechanism.

Use this sheet after a lesson, then cover the answers and explain each connection aloud.

01 KEEP BLOOD AND TUBULAR FLUID SEPARATE

Blood: afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries / vasa recta.

Filtrate: Bowman space → proximal tubule → loop → distal tubule → connecting tubule → collecting system.

Excretion = filtration - reabsorption + secretion.

02 CLEARANCE: ALWAYS CHECK THE UNITS

C = $U \times V / P$ (mL/min). Match urine and plasma concentration units.

Filtered load = $GFR \times P$. Excretion rate = $U \times V$.

FF = GFR / RPF . Example: $120 / 600 = 0.20 = 20\%$.

RBF = $RPF / (1 - \text{hematocrit})$. Hematocrit is a fraction, not a percentage.

Inulin clearance = GFR under ideal conditions. Creatinine secretion makes measured creatinine clearance slightly overestimate GFR. Low-concentration PAH clearance approximates **effective** RPF; extraction is incomplete.

03 FLOW IS NOT THE SAME AS FILTRATION

Afferent constriction: renal blood flow falls; glomerular pressure and GFR tend to fall.

Moderate efferent constriction: plasma flow falls; upstream pressure can support or raise GFR.

Severe efferent constriction: very low plasma flow and increasing capillary oncotic pressure can lower GFR.

Myogenic stretch and high macula-densa NaCl delivery increase afferent tone.

04 KNOW THE TRANSPORT LOCATION

Proximal tubule: bulk, approximately isosmotic sodium and water recovery; most bicarbonate and normally almost all glucose and amino acids.

Thick ascending limb: NKCC2, salt recovery with low water permeability.

Early distal tubule: NCC.

Late distal / collecting system: ENaC, regulated potassium secretion and water handling. Basolateral Na⁺/K⁺-ATPase supports transport.

Recall: Why can PAH clearance exceed inulin clearance? Why does a clearance above GFR not prove dilute urine?

Balance water. Interpret acid-base.

Separate the driving force from the response. A number is useful only when you can explain it.

05 BUILD, PRESERVE, USE THE GRADIENT

Build: loop countercurrent multiplication; urea contributes to the inner medulla.

Preserve: vasa recta countercurrent exchange limits solute washout.

Use: ADH → basolateral V2 receptor → cAMP / PKA → apical AQP2 insertion.

Water follows the osmotic gradient. ADH raises permeability; it does not pump water.

06 VOLUME AND CONCENTRATION ARE DIFFERENT

Extracellular volume relates to total sodium content; osmolality describes solute relative to water. Plasma sodium concentration does not directly measure sodium stores.

Renin: low perfusion, low distal NaCl delivery and sympathetic beta-1 stimulation.

Aldosterone: distal sodium recovery and potassium secretion.

ADH: regulated water permeability; responds to osmolality and substantial low-volume signals.

07 RECLAMATION VERSUS NEW BICARBONATE

Reclaiming filtered bicarbonate prevents its loss. New bicarbonate is associated with net acid excretion.

Net acid excretion = NH_4^+ + titratable acid - urinary bicarbonate.

Alpha intercalated cells secrete H^+ ; beta intercalated cells can secrete bicarbonate.

08 ABG: DIRECTION → PROCESS → CHECK

1. pH: below 7.35 = acidemia; above 7.45 = alkalemia. A pH within range does not exclude mixed processes.

2. Process: low bicarbonate supports metabolic acidosis; high CO_2 supports respiratory acidosis. Opposite changes support alkalosis.

3. Compensation: in metabolic acidosis, expected $\text{PaCO}_2 = 1.5 \times \text{HCO}_3^- + 8 \pm 2$ mmHg. Above the range suggests added respiratory acidosis; below suggests added respiratory alkalosis.

4. Anion gap: $\text{Na}^+ - (\text{Cl}^- + \text{HCO}_3^-)$. Interpret with the local laboratory range, albumin and context.

Example: HCO_3^- 12 predicts PaCO_2 24-28 mmHg. A measured 40 is not adequate respiratory compensation.

Sources and scope. Adapted from the studio's Renal Physiology guide, Revision 15 (2 September 2026), chapters 28-33. Compensation cross-checked against Merck Manual, Acid-Base Disorders. AI-assisted educational adaptation; no independent clinical peer review is claimed. This sheet is not patient-care guidance.

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Source links, 30 explained questions and interactive practice are available in the course.